19-0706; Rev 1; 3/07

MAX16819/MAX16820

2MHz高亮度LED驱动器，具有高边电流检测和5000:1调光范围

概述

MAX16819/MAX16820是降压型、恒流、高亮度LED (HB LED)驱动器，为汽车内部/外部照明、建筑物和环境照明、MR16等LED灯及其它LED照明应用提供极具成效的解决方案。

MAX16819/MAX16820工作在4.5V至28V输入电压范围，内置一个5V/10mA稳压器。输出电流由高边电流检测电阻设置，专用PWM输入(DIM)可实现宽范围的脉冲亮度调节。

MAX16819/MAX16820非常适合需要宽输入电压范围的应用，高边电流检测和内部电流设置电路大大降低了外部元件数量，LED电流精度保持在±5%。在负载瞬变和PWM亮度调节过程中，滞回控制架构可确保优异的输入电源抑制和快速响应。MAX16819具有30%的电感纹波电流，而MAX16820具有10%的纹波电流。这些器件工作在高达2MHz的开关频率，允许使用小尺寸元件。

MAX16819/MAX16820工作于-40°C至+125°C汽车级温度范围，提供3mm x 3mm x 0.8mm、6引脚TDFN封装。

应用

建筑、工业及环境照明

汽车RCL、DRL和雾灯

MR16及其它LED灯

指示器与紧急事件照明灯

特性

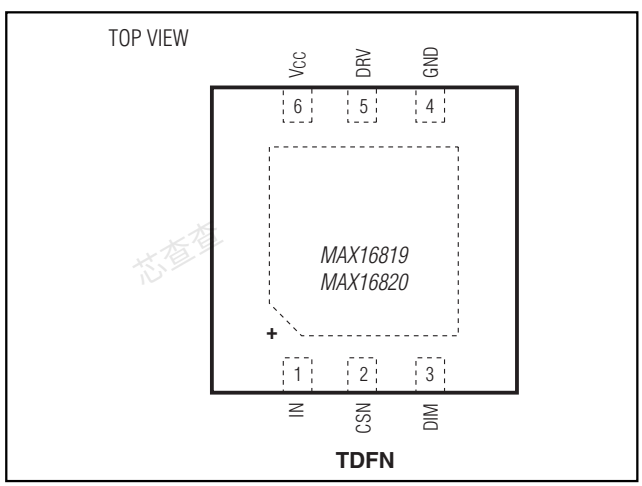
- 高边电流检测
- 专用亮度调节控制输入
- 高达20kHz的亮度调节频率
- 滞回控制：无需补偿
- 开关频率高达2MHz
- LED电流精度：±5%
- 可调节恒定LED电流
- 输入电压范围为4.5V至28V
- 输出功率大于25W
- 5V、10mA内部稳压器
- -40°C至+125°C工作温度范围

定购信息

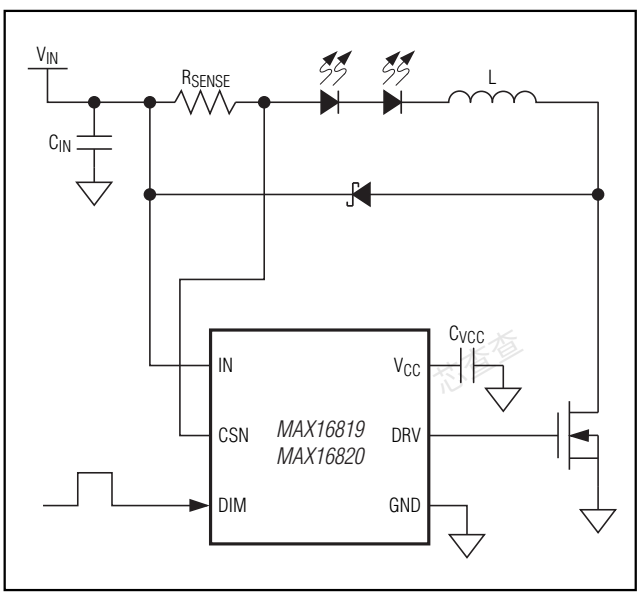
PART	PIN-PACKAGE	PKG CODE	TOP MARK
MAX16819ATT+T	6 TDFN–EP*	T633-2	+ATB
MAX16820ATT+T	6 TDFN-EP*	T633-2	+ATC

注：所有器件工作在-40°C至+125°C温度范围。
+表示无铅封装。
**EP = 裸焊盘。*

引脚配置



典型应用电路



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有关价格、供货及订购信息，请联络Maxim亚洲销售中心：10800 852 1249 (北中国区)，10800 152 1249 (南中国区)，或访问Maxim的中文网站：china.maximintegrated.com。

ABSOLUTE MAXIMUM RATINGS

IN, CSN, DIM to GND .....-0.3V to +30V
V_{CC}, DRV to GND .....-0.3V to +6V
CSN to IN.....-0.3V to +0.3V
Maximum Current into Any Pin
(except IN, V_{CC}, and DRV).....±20mA
Continuous Power Dissipation (T_A = +70°C)
6-Pin TDFN (derate 18.17mW/°C* above +70°C).....1454mW
Operating Temperature Range .....-40°C to +125°C
Junction Temperature.....+150°C
Storage Temperature Range .....-65°C to +150°C
Lead Temperature (soldering, 10s) .....+300°C
Pin-to-Pin ESD Ratings (HB Model).....2.5kV
*As per JEDEC51 Standard (Single-Layer Board).

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

ELECTRICAL CHARACTERISTICS

(V_{IN} = 12V, V_{DIM} = V_{IN}, C_{VCC} = 1µF, R_{SENSE} = 0.5Ω, T_A = T_J = -40°C to +125°C, unless otherwise noted. Typical values are at T_A = +25°C.) (Note 1)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Input Voltage Range	V_{IN}		4.5		28.0	V
Maximum Current Regulator Switching Frequency	f_{SW}				2	MHz
Ground Current	I_{GND}	DRV open			1.5	mA
Supply Current	I_{IN}	V_{DIM} < 0.6V			425	µA
Undervoltage Lockout	UVLO	V_{IN} = V_{CSN} = V_{DIM}, V_{IN} rising from 4V until V_{DRV} > V_{CC} - 0.5V		4.7	5.0	V
		V_{IN} = V_{CSN} = V_{DIM}, V_{IN} falling from 6V, V_{DRV} < 0.5V			4.5	
Undervoltage Lockout Hysteresis				0.5		V
SENSE COMPARATOR						
Sense Voltage Threshold High	V_{SNSHI}	(V_{IN} - V_{CSN}) rising from 0V until V_{DRV} < 0.5V (MAX16820)	195	210	225	mV
		(V_{IN} - V_{CSN}) rising from 0V until V_{DRV} < 0.5V (MAX16819)	213	230	246	
Sense Voltage Threshold Low	V_{SNSLO}	(V_{IN} - V_{CSN}) falling from 0.26V until V_{DRV} > (V_{CC} - 0.5V) (MAX16820)	176	190	204	mV
		(V_{IN} - V_{CSN}) falling from 0.26V until V_{DRV} > (V_{CC} - 0.5V) (MAX16819)	158	170	182	
Propagation Delay to Output High	t_{DPDH}	Falling edge of (V_{IN} - V_{CSN}) from 0.26V to 0V to DRV high, C_{DRV} = 1nF		82		ns
Propagation Delay to Output Low	t_{DPDL}	Rising edge of (V_{IN} - V_{CSN}) from 0V to 0.26V to DRV low, C_{DRV} = 1nF		82		ns
Current-Sense Input Current	I_{CSN}	(V_{IN} - V_{CSN}) = 200mV			1	µA
Current-Sense Threshold Hysteresis	CS_{HYS}	MAX16819		56	70	mV
		MAX16820		17	35	mV

ELECTRICAL CHARACTERISTICS (continued)

(V_{IN} = 12V, V_{DIM} = V_{IN}, C_{VCC} = 1µF, R_{SENSE} = 0.5Ω, T_A = T_J = -40°C to +125°C, unless otherwise noted. Typical values are at T_A = +25°C.) (Note 1)

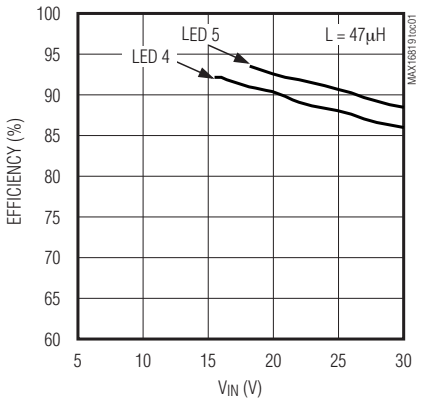
PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
GATE DRIVER						
Gate Driver Source Current		V_{CSN} = V_{IN}, V_{DRV} = 0.5 x V_{CC}		0.5		A
Gate Driver Sink Current		V_{CSN} = V_{IN} - 250mV, V_{DRV} = 0.5 x V_{CC}		1		A
Gate Driver Output-Voltage High	V_{OH}	I_{DRV} = 10mA	V_{CC} - 0.5			V
Gate Driver Output-Voltage Low	V_{OL}	I_{DRV} = -10mA			0.5	V
DIM INPUT						
Maximum DIM Frequency	f_{DIM}				20	kHz
DIM Input-Voltage High	V_{IH}	V_{CSN} = V_{IN}, increase DIM until V_{DRV} > (V_{CC} - 0.5V)	2.8			V
DIM Input-Voltage Low	V_{IL}	V_{CSN} = V_{IN}, decrease DIM until V_{DRV} < 0.5V			0.6	V
DIM Hysteresis	DIM_{HYS}			200		mV
DIM Turn-On Time	t_{DIMON}	DIM rising edge to V_{DRV} = 0.5 x V_{CC}, C_{DRV} = 1nF		100		ns
DIM Turn-Off Time	t_{DIMOFF}	DIM falling edge to V_{DRV} = 0.5 x V_{CC}, C_{DRV} = 1nF		100		ns
DIM Input Leakage High		V_{DIM} = V_{IN}			10	µA
DIM Input Leakage Low		V_{DIM} = 0V	-1		+1	µA
V_{CC} REGULATOR						
Regulator Output Voltage	V_{CC}	I_{VCC} = 0.1mA to 10mA, V_{IN} = 5.5V to 28V	4.5		5.5	V
		I_{VCC} = 0.1mA to 10mA, V_{IN} = 4.5V to 28V	4.0		5.5	V
Load Regulation		I_{VCC} = 0.1mA to 10mA, V_{IN} = 12V		4		Ω
Line Regulation		V_{IN} = 6V to 28V, I_{VCC} = 10mA		11		mV
Power-Supply Rejection Ratio	PSRR	V_{IN} = 12V, I_{VCC} = 5mA, f_{IN} = 10kHz		-35		dB
Current Limit	I_{LIM}	V_{IN} = 4.5V, V_{CC} = 0V		45		mA
		V_{IN} = 4.5V, V_{CC} = 4V		18		mA
Regulator Startup Time	t_{STRAT}	V_{CC} = 0 to 4.5V		350		µs

Note 1: All devices are 100% production tested at T_J = +25°C and +125°C. Limits to -40°C are guaranteed by design.

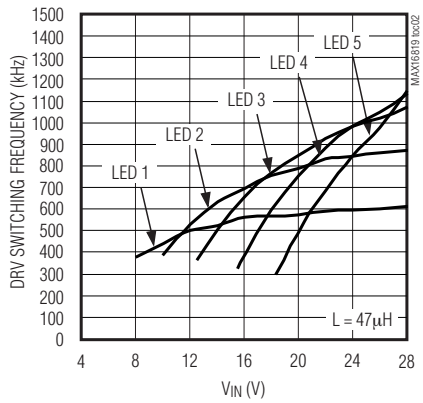
典型工作特性

(V_{IN} = V_{DIM} = 12V, C_{VCC} = 1µF, R_{SENSE} = 0.5Ω connected between IN and CSN. Typical values at T_A = +25°C, unless otherwise noted.)

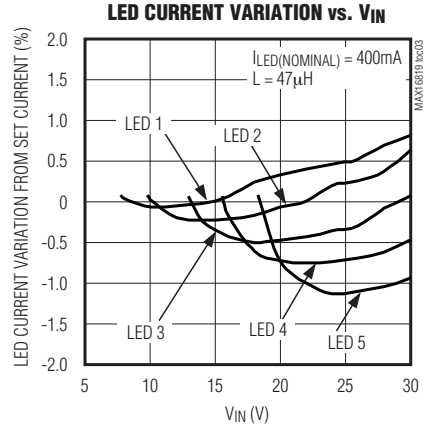
EFFICIENCY vs. V_{IN}



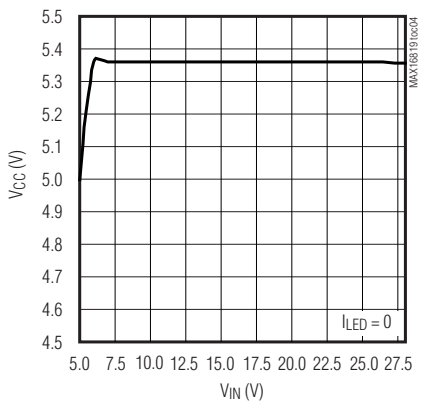
DRV SWITCHING FREQUENCY vs. V_{IN}



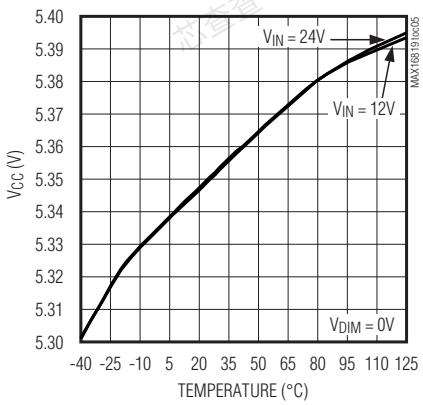
LED CURRENT VARIATION vs. V_{IN}



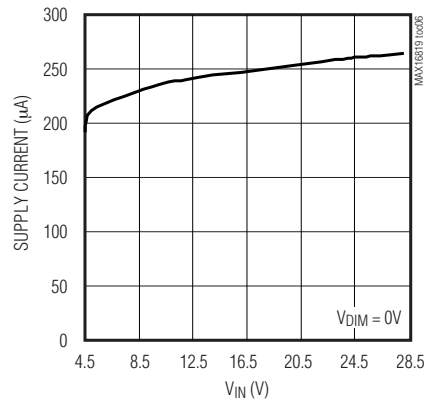
V_{CC} vs. V_{IN}



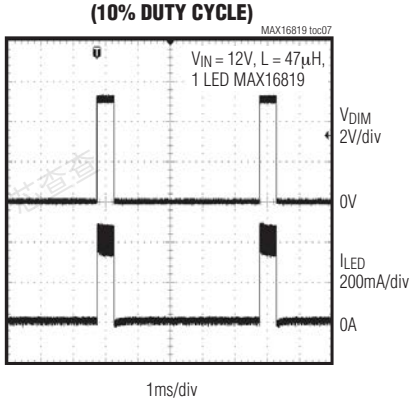
V_{CC} vs. TEMPERATURE



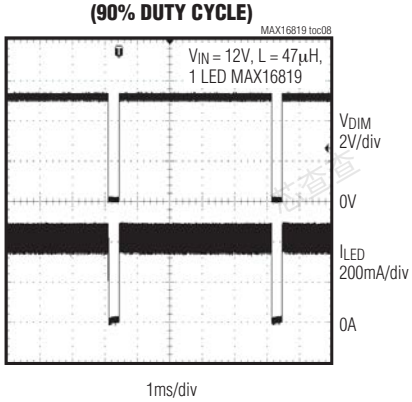
SUPPLY CURRENT vs. V_{IN}



PWM DIMMING AT 200Hz (10% DUTY CYCLE)



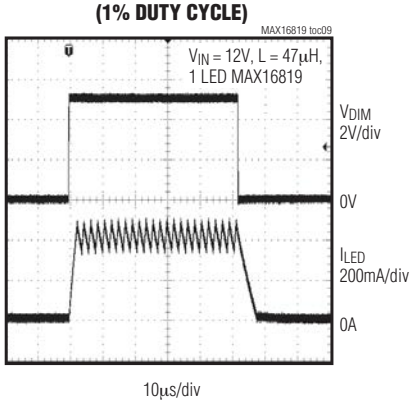
PWM DIMMING AT 200Hz (90% DUTY CYCLE)



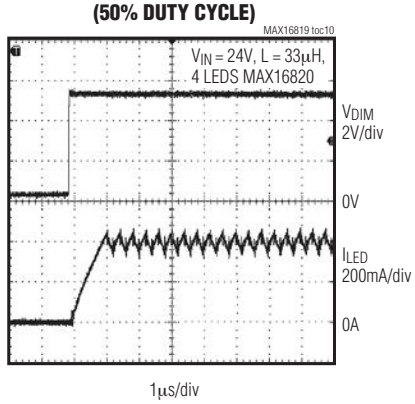
典型工作特性(续)

(V_{IN} = V_{DIM} = 12V, C_{VCC} = 1µF, R_{SENSE} = 0.5Ω connected between IN and CSN. Typical values at T_A = +25°C, unless otherwise noted.)

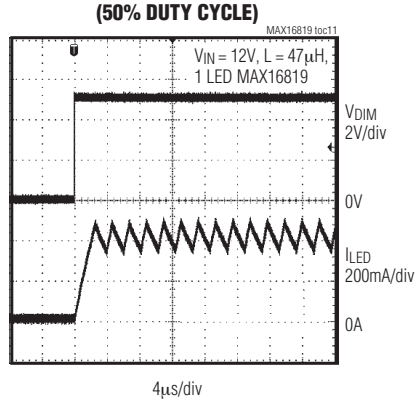
PWM DIMMING AT 200Hz (1% DUTY CYCLE)



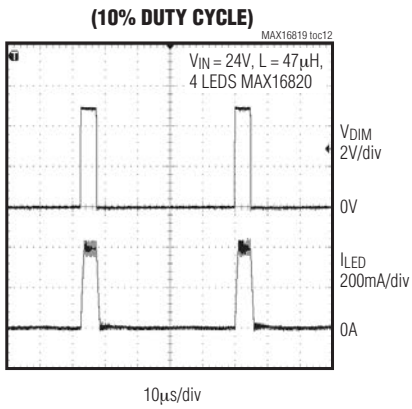
PWM DIMMING EXPANDED (50% DUTY CYCLE)



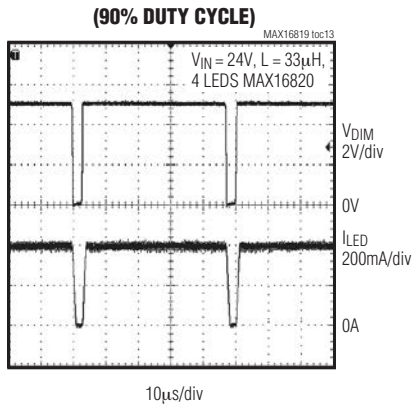
PWM DIMMING EXPANDED (50% DUTY CYCLE)



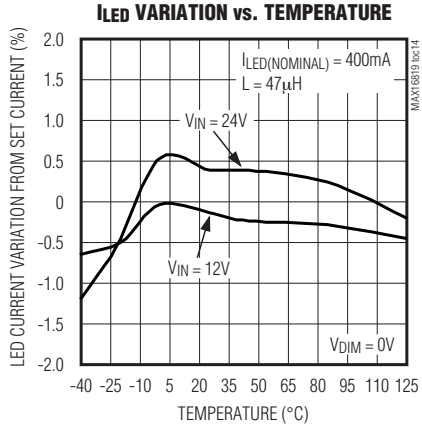
PWM DIMMING AT 20kHz (10% DUTY CYCLE)



PWM DIMMING AT 20kHz (90% DUTY CYCLE)



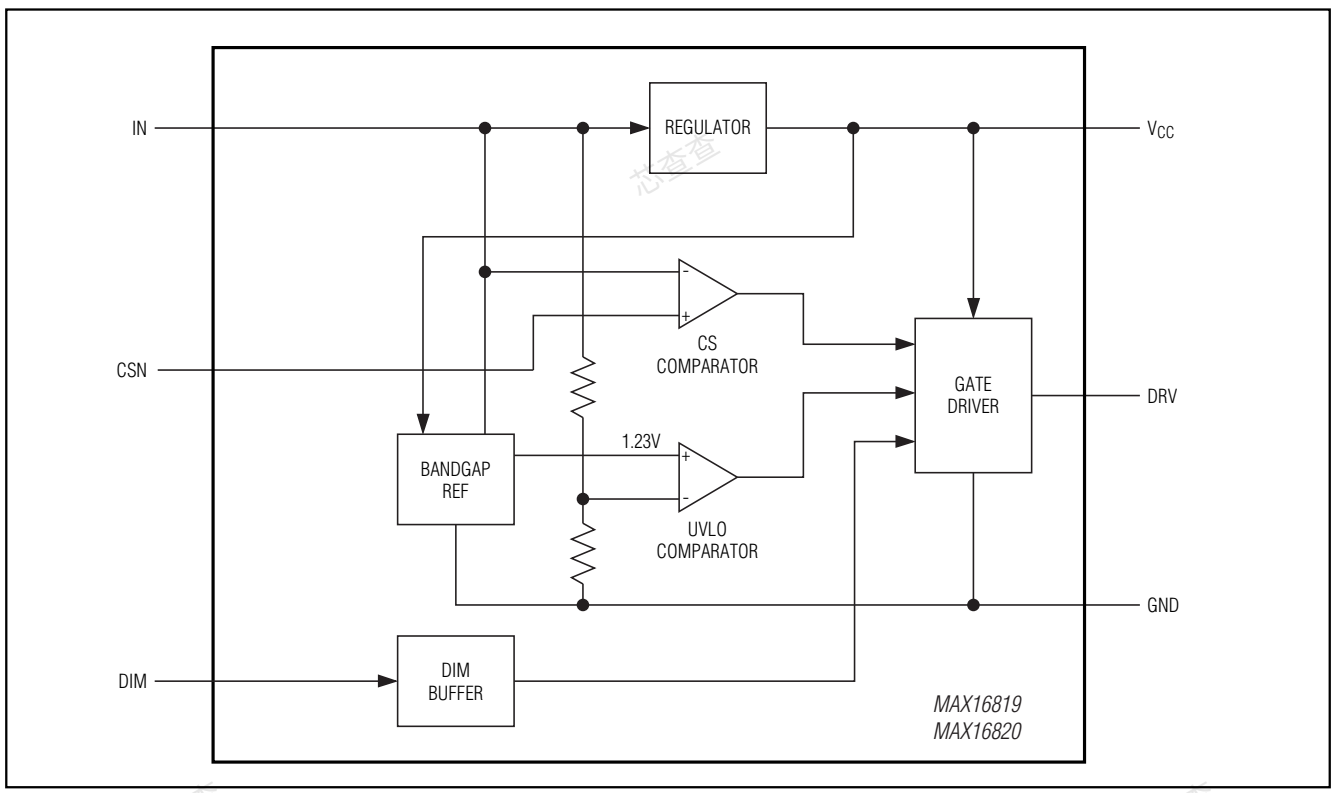
I_{LED} VARIATION vs. TEMPERATURE



引脚说明

引脚	名称	功能
1	IN	正电源输入，通过一个1µF或更高容值的电容旁路至GND。
2	CSN	电流检测输入。
3	DIM	逻辑电平调光输入，驱动DIM为低电平将关断电流调节器；驱动DIM为高电平则开启电流调节器。
4	GND	地。
5	DRV	栅极驱动输出，连接至外部n沟道MOSFET的栅极。
6	V_{CC}	稳压器输出，用一个1µF电容旁路V_{CC}至GND。
EP	—	裸焊盘，连接至大面积地层以改善散热条件，不要作为器件的唯一接地点。

功能框图



详细说明

MAX16819/MAX16820是降压型、恒流、高亮度LED (HB LED)驱动器，工作于4.5V至28V输入电压范围，能够为外部MOSFET的栅极提供高达0.5A的源出电流和1A的灌入电流。通过高边电流检测电阻设置输出电流，专用的PWM亮度调节输入(DIM)能够支持较宽的脉冲调光范围。

高边电流检测配置和内部电流设置电路大大减少了外部元件数量，使用1%的检测电阻时具有±5%的LED电流精度，请参考*功能框图*。

欠压锁定(UVLO)

MA16819/MAX16820包含4.5V欠压锁定(UVLO)电路，具有500mV的滞回电压。当V_{IN}降至4.5V以下时，DRV变为低电平，关断外部n沟道MOSFET。一旦V_{IN}达到5V或更高电压，DRV将变为高电平。

5V稳压器

V_{CC}是5V稳压器输出，可提供10mA电流。利用1µF电容旁路V_{CC}至GND。

DIM输入

MAX16819/MAX16820可通过DIM输入的PWM信号进行亮度调节。DIM端的逻辑电平低于0.6V时，强制MAX16819/MAX16820的DRV输出低电平，从而关断LED电流。开通LED电流时，DIM端的逻辑电平必须至少为2.8V。

应用信息

选择R_{SENSE}设定LED电流

MAX16819/MAX16820可利用连接在IN和CSN之间的电阻对LED电流进行编程设置，通过下式计算检流电阻：

$$R_{SENSE}(\Omega) = \frac{1}{2}\frac{(V_{SNSHI} + V_{SNSLO})(V)}{I_{LED}(A)}$$

关于V_{SNSHI}和V_{SNSLO}的说明，请参考*Electrical Characteristics*表。

电流调节

MAX16819/MAX16820利用一个滞回比较器(图1)调节LED输出电流。随着电感电流上升，使检流电阻两端的电压达到门限电压的上限时，DRV输出低电平，关断外部MOSFET。当电感电流通过续流二极管下降，并使检流电阻两端的电压达到门限电压的下限时，MOSFET才重新导通。利用下式确定工作频率：

$$f_{SW} = \frac{(V_{IN} - n \times V_{LED}) \times n \times V_{LED} \times R_{SENSE}}{V_{IN} \times \Delta V \times L}$$

其中n = LED的数量，V_{LED} = 单个LED的正向压降，$\Delta V = (V_{SNSHI} - V_{SNSLO})$。

请利用设计工具：http://www.maxim-ic.com.cn/MAX16819-20-Tool正确选择元件。

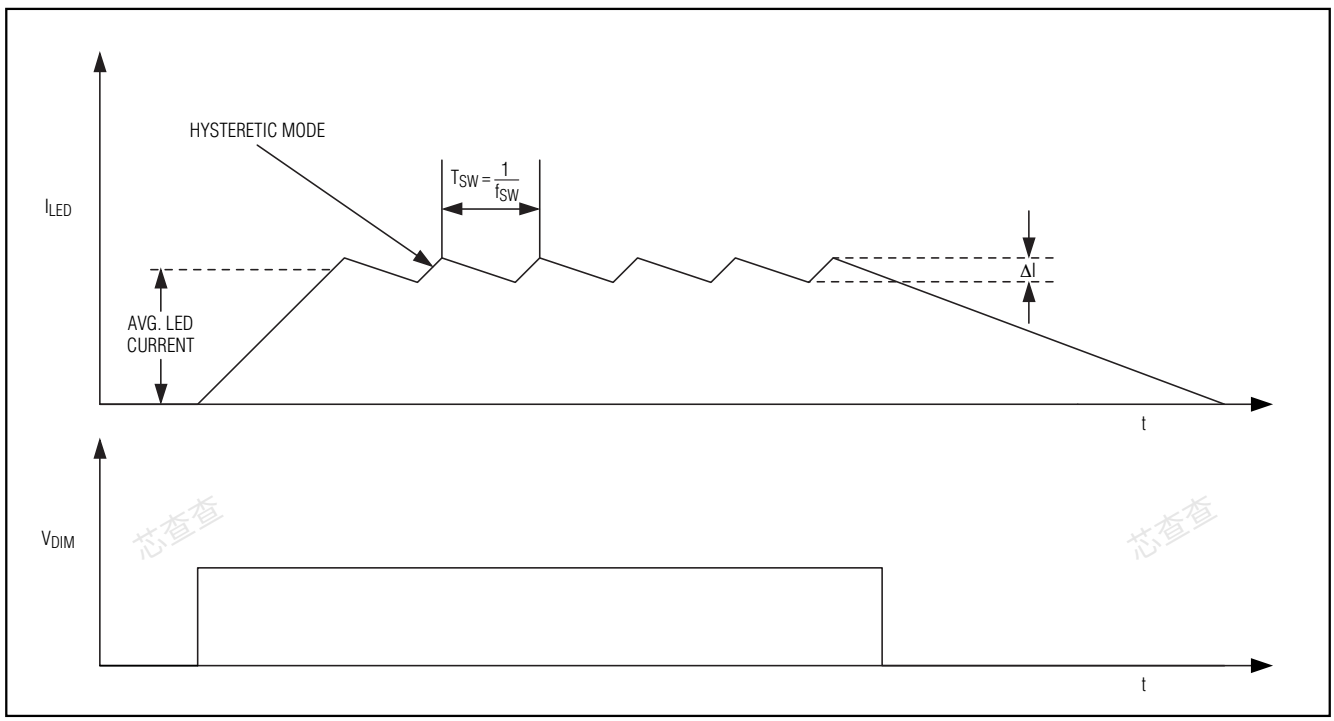


图1. 电流调节器

MOSFET选择

MAX16819/MAX16820栅极驱动器可提供0.5A的输出电流和1A灌入电流，应根据最大输入电压V_{IN}、输出电流I_{LED}和开关频率选择MOSFET。选择击穿电压高于最大工作电压、具有低$R_{DS(ON)}$及低电荷的MOSFET，以获得更佳效率。如果工作在输入电压范围的最小值，须注意MOSFET的门限电压必须满足要求。

续流二极管的选择

续流二极管的正向导通压降应尽可能低，以获得更佳效率。只要击穿电压足以承受最大工作电压，可优先选择肖特基二极管。

二极管的正向额定电流不能低于最大LED电流。

LED纹波电流

LED纹波电流等于电感的纹波电流，要求较低的LED纹波电流时，可在LED端口接一个电容。

PCB布线指南

细致的PCB布线对获得低开关损耗和稳定的工作状态至关重要，尽可能使用多层板以便更好地抑制噪声干扰。大电流接地回路、输入旁路电容地线和输出滤波器地线采用单点连接(星形接地结构)，进一步降低接地噪声。正常工作状态下有两个大功率回路：一个回路是MOSFET导通时的大电流回路，由IN—R_{SENSE}—LED—电感—MOSFET—GND形成；另一个回路是MOSFET关断时的大电流回路，由R_{SENSE}—LED—电感—续流二极管形成。为降低噪声干扰，每个回路的面积应尽可能小。

R_{SENSE}须尽可能靠近输入滤波器和IN引脚放置，为提高抗干扰能力，建议CSN和R_{SENSE}采用Kelvin连接。裸焊盘接至大面积地层，以提高散热能力。

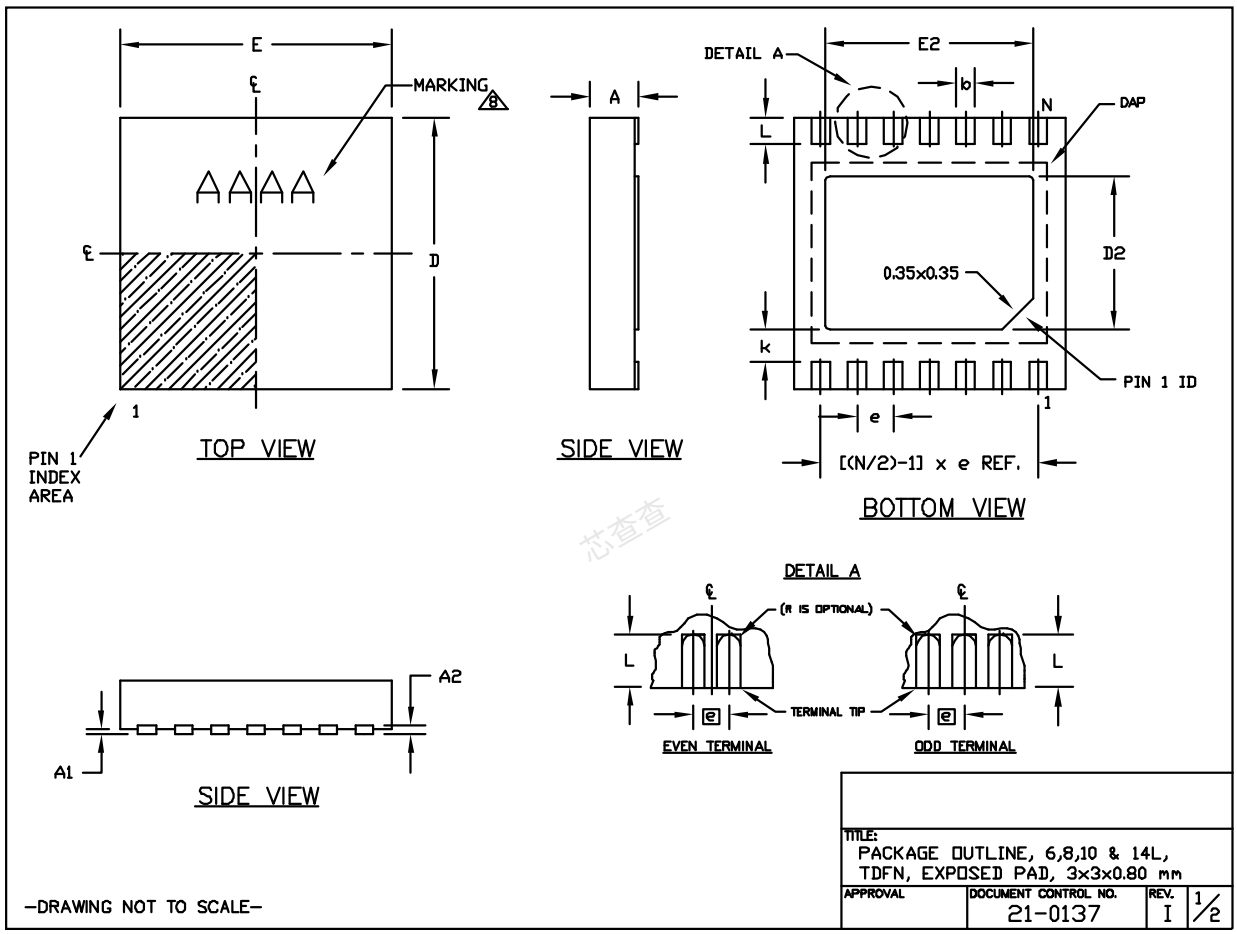
芯片信息

PROCESS: BiCMOS

芯查查

封装信息

(本数据资料提供的封装图可能不是最近的规格，如需最近的封装外形信息，请查询 www.maxim-ic.com.cn/packages.)



封装信息(续)

(本数据资料提供的封装图可能不是最近的规格，如需最近的封装外形信息，请查询 www.maxim-ic.com.cn/packages。)

COMMON DIMENSIONS		
SYMBOL	MIN.	MAX.
A	0.70	0.80
D	2.90	3.10
E	2.90	3.10
A1	0.00	0.05
L	0.20	0.40
k	0.25 MIN.	
A2	0.20 REF.	

PACKAGE VARIATIONS							
PKG. CODE	N	D2	E2	e	JEDEC SPEC	b	[(N/2)-1] x e
T633-2	6	1.50±0.10	2.30±0.10	0.95 BSC	MO229 / WEEA	0.40±0.05	1.90 REF
T833-2	8	1.50±0.10	2.30±0.10	0.65 BSC	MO229 / WEEC	0.30±0.05	1.95 REF
T833-3	8	1.50±0.10	2.30±0.10	0.65 BSC	MO229 / WEEC	0.30±0.05	1.95 REF
T1033-1	10	1.50±0.10	2.30±0.10	0.50 BSC	MO229 / WEED-3	0.25±0.05	2.00 REF
T1033-2	10	1.50±0.10	2.30±0.10	0.50 BSC	MO229 / WEED-3	0.25±0.05	2.00 REF
T1433-1	14	1.70±0.10	2.30±0.10	0.40 BSC	- - - -	0.20±0.05	2.40 REF
T1433-2	14	1.70±0.10	2.30±0.10	0.40 BSC	- - - -	0.20±0.05	2.40 REF

NOTES:
1. ALL DIMENSIONS ARE IN mm. ANGLES IN DEGREES.
2. COPLANARITY SHALL NOT EXCEED 0.08 mm.
3. WARPAGE SHALL NOT EXCEED 0.10 mm.
4. PACKAGE LENGTH/PACKAGE WIDTH ARE CONSIDERED AS SPECIAL CHARACTERISTIC(S).
5. DRAWING CONFORMS TO JEDEC MO229, EXCEPT DIMENSIONS "D2" AND "E2", AND T1433-1 & T1433-2.
6. "N" IS THE TOTAL NUMBER OF LEADS.
7. NUMBER OF LEADS SHOWN ARE FOR REFERENCE ONLY.
8. MARKING IS FOR PACKAGE ORIENTATION REFERENCE ONLY.

-DRAWING NOT TO SCALE-

TITLE: PACKAGE OUTLINE, 6,8,10 & 14L, TDFN, EXPOSED PAD, 3x3x0.80 mm			
APPROVAL	DOCUMENT CONTROL NO. 21-0137	REV. I	2/2

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修订历史

Rev 1中的修改页：1、4、10。

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